Muscle Sympathetic Nerve Activity Burst Identification and Analysis Standard Operating Procedure

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Glossary

a. **Continuous Files:** files that do not have segments or intervention periods. For example, a 10-min baseline resting file.

b. **CPT:** cold-pressor test

c. **ECG**: electrocardiogram

d. **EPR:** exercise-pressor reflex

e. **MSNA**: muscle sympathetic nerve activity

f. **NIBP:** non-invasive blood pressure

g. **Segmented Files:** files that have segments or intervention periods. For example, an exercise pressor reflex testing file has four segments (baseline, exercise, post-exercise circulatory occlusion, and recovery).

LabChart Setup

IMPORTANT NOTE: It will take time to initially set up this program for analysis, but once all components are created, all subsequent analyses will be very quick. This section will detail, from the beginning, how to analyze continuous baseline files. This program will facilitate the creation of just one workbook for muscle sympathetic nerve activity (MSNA) data per participant to analyze all MSNA data for each study. It is suggested to create a template workbook for each study, so this workbook can be duplicated for each new participant.

Cut Files and Prepping Macros

*****LabChart macros currently do not work on Macintosh systems, limiting these to Windows systems.**

Setting up LabChart Audio Playback: If this feature is not already installed in LabChart, follow the link below to install Audio Output on LabChart. LabChart may need to be restarted after installation.
https://www.adinstruments.com/products/audio-output?srsltid=AfmBOopGR78B3IYwoWMSAYGGv2qMl2V1P5rFDrUCBgJGzvx-CF-UdrDf

Cutting Files

I. Create separate LabChart files for each study procedure of interest. For example, one LabChart file will be baseline rest, another LabChart file will be, for example, a cold-pressor test file.

    A. For all continuous files (baseline rest, or any files that do not have separate sections such as baseline, stressor, recovery), the file should be highlighted from comment to comment. Next, *File → Save Selection* should be selected, and the selection saved in the desired folder. Reference the ADInstruments website for additional help with this step.

    B. For all segmented files that have separate sections such as baseline, stressor, recovery, the file should be highlighted from the FIRST comment (for example, Baseline CPT [cold pressor test]) up until the final comment following recovery if that data is of interest (for example, CPT Recovery End). Next, *File → Save Selection* should be selected, and the selection saved in the desired folder. Again, all files should be kept structured and organized.

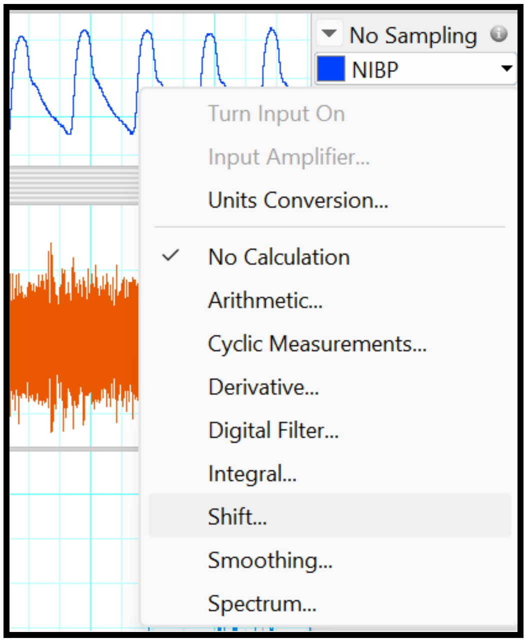
Prepping LabChart Cut Files

I. Open the LabChart file that will be analyzed (e.g., SUBID_Rest or SUBID_CPT, etc.).

II. The time scale should be changed to seconds from the start of the file: ***right-click the area at the bottom near the time stamps → Time Mode → select "Show time from start of file" → select "Show time as seconds"***. At the beginning of the file, the time scale should read 0, and the end should read the total? file duration in seconds.

III. All existing comments should be deleted. If all existing comments are not deleted, downstream errors in the analysis may occur.
IMPORTANT NOTE: If segmented MSNA analysis is being performed (analysis during stressors), the length of each segment (i.e., baseline, CPT, CPT recovery) should be recorded in a separate document ***BEFORE*** all comments are deleted. This information will be needed in order to appropriately perform the MSNA analysis in the workbook.

IV. Ensure that the peak of the non-invasive blood pressure (NIBP) waveform is aligned with the electrocardiogram (ECG) peak to ensure that the appropriate diastolic blood pressure (BP) value falls within the correct cardiac cycle. ***Click on the dropdown menu shown in the image below → Shift → shift as necessary depending on the hardware setup***. Also note that different hardware requires different time shifts. For example, if using the reconstructed brachial BP on a Finapres Nova Device, the negative shift is 4s, since the device requires 4s before the reconstructed BP is calculated and sent to LabChart.



The next steps (VI to XIV) will detail how to set up macros so that these steps do not need to be manually followed with every participant file.

V. __**Channel Setup:**__

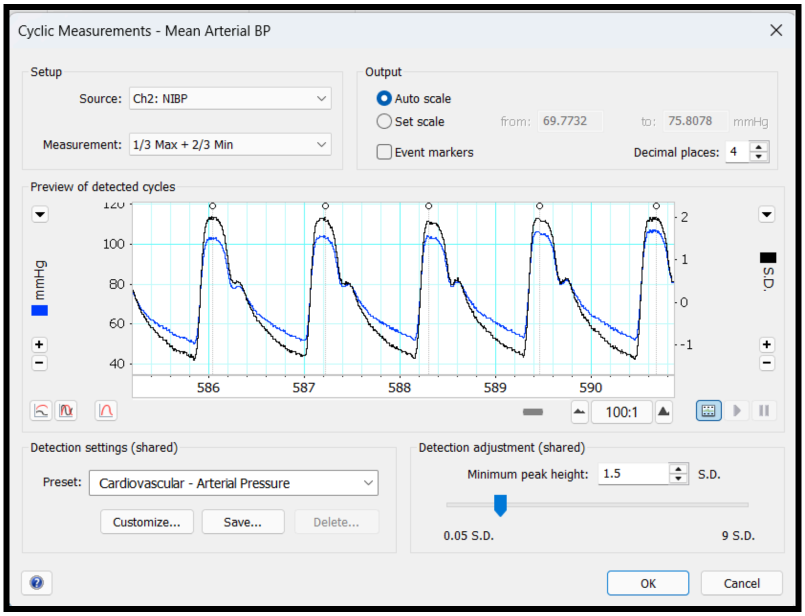
*__**IMPORTANT NOTE**__*: when creating these macros it is suggested to record the steps that are being completed and label the Macro as Channel Setup. If macros are recorded, then they only need to be set up once and they can be uploaded into subsequent files.
To create a macro: ***Macro → Manage → New → Name Macro, Show in toolbar → Click Record***

ECG: A visible ECG signal is needed for BEAT comments to be placed (next steps).

NIBP: Depending on the hardware, as previously stated, this signal may need to be shifted.

Integrated/RMS MSNA: Either signal will work in this workflow.

Mean Arterial BP: ***Cyclic Measurements → Preset is Cardiovascular-Arterial Pressure → Measurement 1/3 Max + 2/3 Min→ see image below***



Cyclic Measurements - Mean Arterial BP

Setup

Source: Ch2: NIBP

Measurement: 1/3 Max + 2/3 Min

Output

● Auto scale

○ Set scale from: 69.7732 to: 75.8078 mmHg

☐ Event markers

Decimal places: 4

Preview of detected cycles

Detection settings (shared)

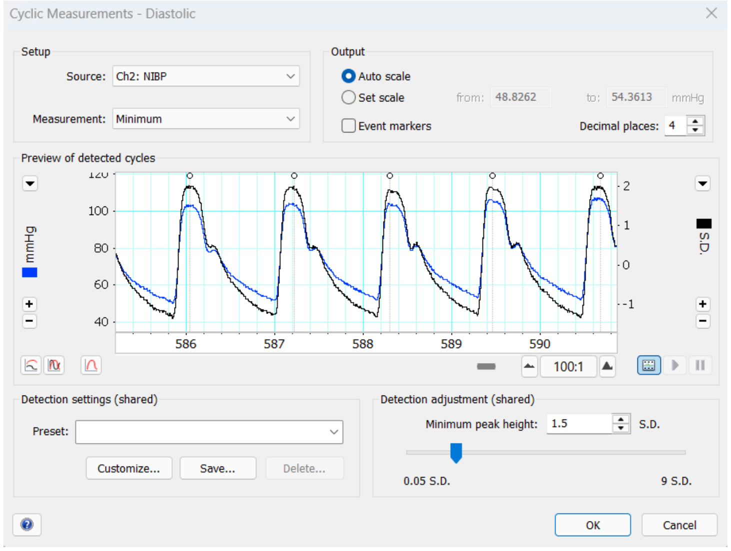
Preset: Cardiovascular - Arterial Pressure

Detection adjustment (shared)

Minimum peak height: 1.5 S.D.

120
100
80
60
40
mmHg
586
587
588
589
590
2
1
0
-1
S.D.
100:1
0.05 S.D.
9 S.D.
Customize...
Save...
Delete...
OK
Cancel

Diastolic BP: ***Cyclic Measurements → Preset is Cardiovascular-Arterial Pressure → Measurement is Minimum→ see image below:***



Cyclic Measurements - Diastolic

Setup

Source: Ch2: NIBP

Measurement: Minimum

Output

● Auto scale

○ Set scale from: 48.8262 to: 54.3613 mmHg

☐ Event markers

Decimal places: 4

Preview of detected cycles

Detection settings (shared)

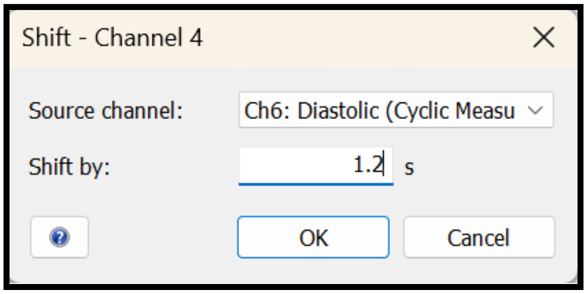
Preset:

Detection adjustment (shared)

Minimum peak height: 1.5 S.D.

120
100
80
60
40
mmHg
586
587
588
589
590
2
1
0
-1
S.D.
100:1
0.05 S.D.
9 S.D.
Customize...
Save...
Delete...
OK
Cancel

Shifted Diastolic BP:
Create a separate channel (right click on screen and press Add Channel or go to Setup → Channel Settings → Increase the # of channels at the bottom of the popup window) → shift → see image below. This channel will be used to calculate sBRS, which incurs a ~1.2s physiological delay, where the burst propagated based on the diastolic BP (DBP) value occurs ~1.2s later. In reality, a range exists here depending on the recording limb, height of the participant, and resting heart rate. This range can span 0.9–1.54s, thus we suggest the researcher confirm an average reflex latency manually using a Selection Duration feature which is described in the following step (**VI. Column J: Selection Duration (s)**). This number may be adjusted as seen fit for a given laboratory. The channel number (Ch6) is dependent on the channel # at which the diastolic BP channel is located.



Source channel: Ch6: Diastolic (Cyclic Measu

Shift by: 1.2 s

→ ***ONCE THESE STEPS ARE COMPLETED PRESS*** *Stop recording and save.*

VI. **Data Pad Setup**: Set up columns in the Data Pad as stated below:
*Tip: record a macro of these steps being completed and label the macro Datapad Setup or something similar.
Macro → Manage → New → Name Macro, Show in toolbar → Click Record

Column A: Integrated MSNA Amplitude:
Statistics → Maximum - Minimum → Source Channel: Integrated/RMS MSNA
******You may swap out burst amplitude for Burst Area for your analyses. If so, use Integral → Integral relative to Baseline → Source Channel: Integrated/RMS MSNA***

Column B: Integrated MSNA Comment Number:
Comments → Comment Number → Source Channel: Integrated/RMS MSNA

Column C: Time (s):
Selection & Active Point → Time

Column D: Integrated MSNA Comment Time (s):
Comments → Comment Time → Source Channel: Integrated/RMS MSNA

Column E: ECG Full Comment Text:
Comments → Full Comment Text → Source Channel: ECG

Column F: ECG Comment Number:
Comments → Comment Number → Source Channel: ECG

Column G: Mean Arterial BP Value (mmHg):
Selection & Active Point → Value → Source Channel: Mean Arterial BP

Column H: Diastolic BP Value (mmHg):
Selection & Active Point → Value → Source Channel: Diastolic BP

Column I: Shifted Diastolic BP Value (mmHg):
Selection & Active Point → Value → Source Channel: Shifted Diastolic BP

Column J: Selection Duration (s):
Selection & Active Point → Selection Duration
***use this feature to confirm reflex latency (see images below): (1) click on the down arrow in Data Pad → Data Pad Miniwindow (**Image 1**), (2) click on J- Selection Duration (s) **(Image 1**), (3) click, drag, and highlight from the peak of a burst to the previous R-wave and note the duration (**Image 2**), (4) repeat across several bursts, (5) the latency you choose to shift the Shifted Diastolic BP channel does not need to be super specific because DBP is calculated across an entire cardiac cycle, but it is suggested to double check that the correct DBP value is aligned in the shifted channel. Essentially for sBRS calculations, the data is exported based on BEAT comments, thus the shifted DBP channel must be aligned with the BEAT that the burst associated with it lays in. Depending on the participant, this may mean shifting the Shifted DBP Channel more than the calculated latency suggests. For example, the burst in **Image 3** is associated with a cardiac cycle whose DBP was 75.8187mmHg with a ~1.3s latency. However, in order to align the correct DBP value with its burst, we had to shift the Shifted DBP Channel by 1.6s.

Image 1

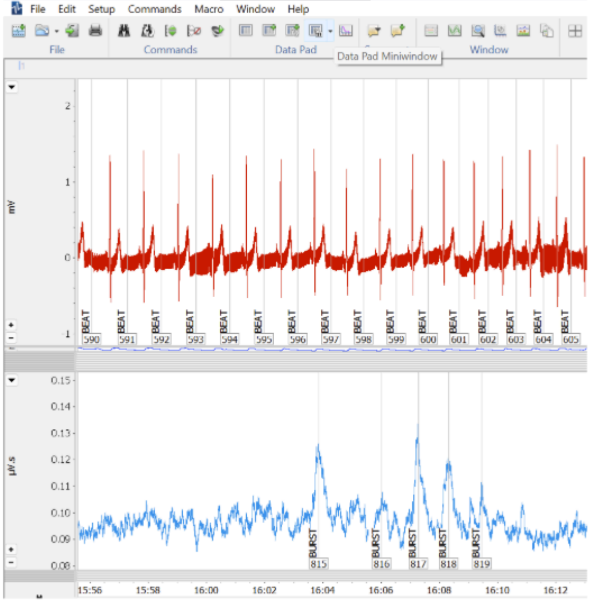
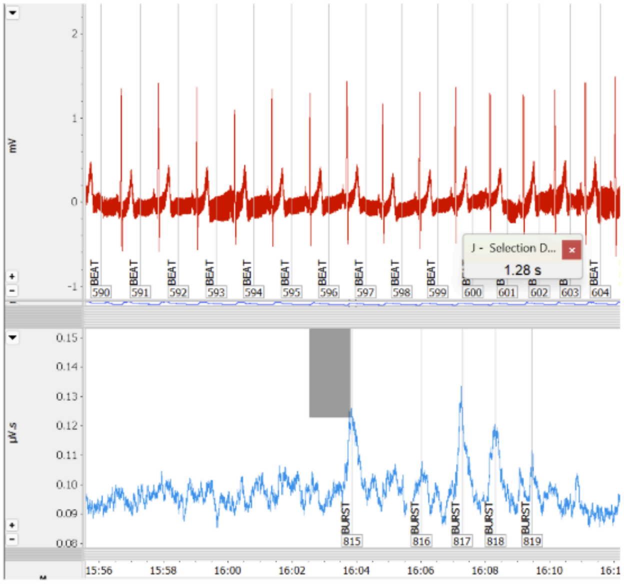

File Edit Setup Commands Macro Window Help
File
Commands
Data Pad
Data Pad Miniwindow
Window
mV
µV.s
BEAT 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605
BURST 815 816 817 818 819
15:56 15:58 16:00 16:02 16:04 16:06 16:08 16:10 16:12

Image 2


mV
µV.s
J - Selection D...
1.28 s
BEAT 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604
BURST 815 816 817 818 819
15:56 15:58 16:00 16:02 16:04 16:06 16:08 16:10 16:1

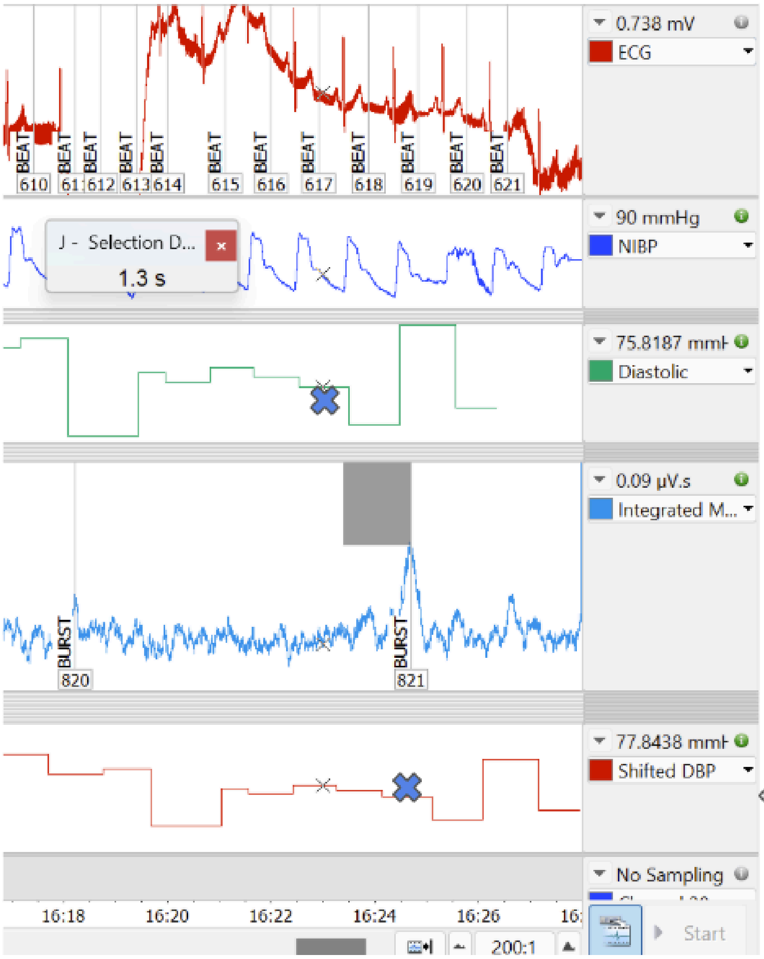
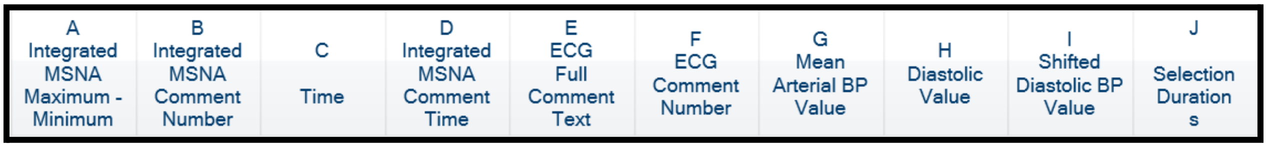

BEAT 610 61 612 613 614 615 616 617 618 619 620 621
0.738 mV
ECG
J - Selection D...
1.3 s
90 mmHg
NIBP
75.8187 mmH
Diastolic
BURST 820 821
0.09 µV.s
Integrated M...
77.8438 mmH
Shifted DBP
No Sampling
16:18 16:20 16:22 16:24 16:26 16:
200:1
Start
Shifted +1.6s

Image #3

Turn OFF all other channels by clicking on the column header and selecting Off. Not necessary, but it keeps the Datapad visually clean and easier to export into Excel.

Data Pad headers should look like the image below:



A	B	C	D	E	F	G	H	I	J
Integrated MSNA Maximum - Minimum	Integrated MSNA Comment Number	Time	Integrated MSNA Comment Time	ECG Full Comment Text	ECG Comment Number	Mean Arterial BP Value	Diastolic Value	Shifted Diastolic BP Value	Selection Durations

*→ **ONCE THESE STEPS ARE COMPLETED PRESS Stop recording and save.***

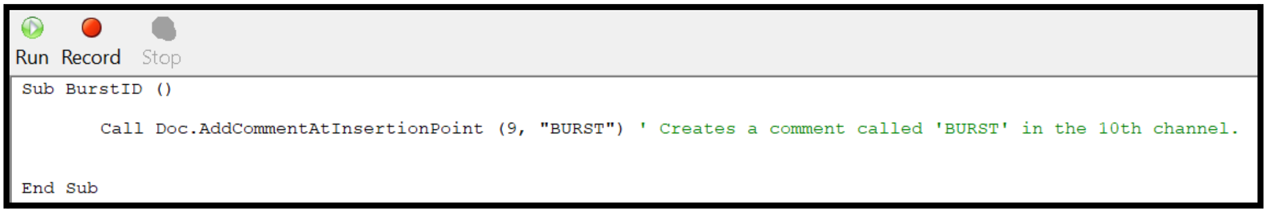
VII. Create Burst ID Macro:

This macro is used to manually mark MSNA bursts.

Macro → Manage → New → Type: Simple Macro, Name: BurstID → Check Show in toolbar → Click OK → Underneath the line Sub Burst ID (), copy the code below:

```
Call Doc.AddCommentAtInsertionPoint (X, "BURST")
```

*For **X**, use the n-1 rule to enter in as Channel # for the Integrated/RMS MSNA Channel. For example, if the Integrated/RMS MSNA Channel is Channel 10, then enter the number 9, and the complete code should read:*



```
Sub BurstID ()

        Call Doc.AddCommentAtInsertionPoint (9, "BURST") ' Creates a comment called 'BURST' in the 10th channel.

End Sub
```

There are automated ways to perform burst marking, which are beyond the scope of the current paper. Please contact the corresponding author if interested in learning about these methods.

VIII. Create a DATAPAD_BEAT Macro:

This macro will look for BEAT comments (these will be placed between each R-R interval in the ECG Channel, and then extract data--the data columns that were set up in the Data Pad-- that are associated with each BEAT comment).

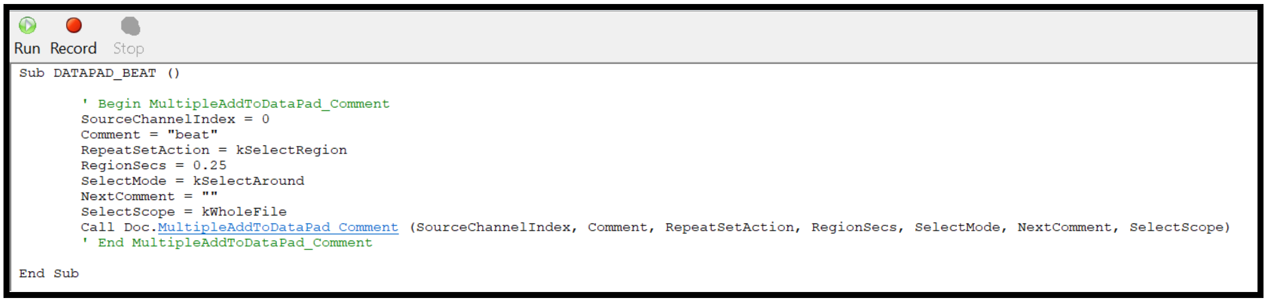
Macro → Manage → New → Type: Simple Macro, Name: DATAPAD_BEAT → Check Show in toolbar → Click OK → Underneath the line Sub DATAPAD_BEAT (), copy the code below:

```
SourceChannelIndex = X
Comment = "beat"
RepeatSetAction = kSelectRegion
RegionSecs = 0.25
SelectMode = kSelectAround
NextComment = ""
SelectScope = kWholeFile
Call Doc.MultipleAddToDataPad_Comment (SourceChannelIndex, Comment, RepeatSetAction, RegionSecs, SelectMode, NextComment, SelectScope)
```

*******For the line of code highlighted blue, when copied over from the PDF into LabChart, it will sometimes break the code into two or more lines. Please make sure the line of code remains on a single line (or as a single sentence without pressing the Enter key) or else it may return an error. There may be other lines of code in the following step that do the same, so if an error is returned make sure to check that lines that are meant to be together, remain together.****

*******RegionSecs may need to be modified depending on the participant's heart rate. This is more affected by high heart rates sustained during exercise. If you notice calculation errors of MSNA outcomes during stressor components, you may need to decrease the RegionSecs to a lower value to capture correct data associated with each cardiac cycle.****

For X, use the n-1 rule to enter in as the Channel # for the ECG Channel. For example, if the ECG Channel is Channel 1, then enter the number 0, and the complete code should read:



```
Sub DATAPAD_BEAT ()

        ' Begin MultipleAddToDataPad_Comment
        SourceChannelIndex = 0
        Comment = "beat"
        RepeatSetAction = kSelectRegion
        RegionSecs = 0.25
        SelectMode = kSelectAround
        NextComment = ""
        SelectScope = kWholeFile
        Call Doc.MultipleAddToDataPad_Comment (SourceChannelIndex, Comment, RepeatSetAction, RegionSecs, SelectMode, NextComment, SelectScope)
        ' End MultipleAddToDataPad_Comment

End Sub
```

IX. **Create a DATAPAD_BURST Macro:**

This macro will look for BURST comments (the comments made when bursts are manually/automatically marked in the Integrated/RMS MSNA channel). Data (i.e., the data columns set up in the Data Pad) associated with each BURST comment will then be extracted. In LabChart, data needs to be exported based on BURST comments separately from BEAT comments, because all bursts in a file cannot be time-aligned with BEAT comments. Thus, in order to preserve data associated with burst amplitude, these data need to be exported separately. This macro will pull data from a 0.5s. search window surrounding each BURST comment.

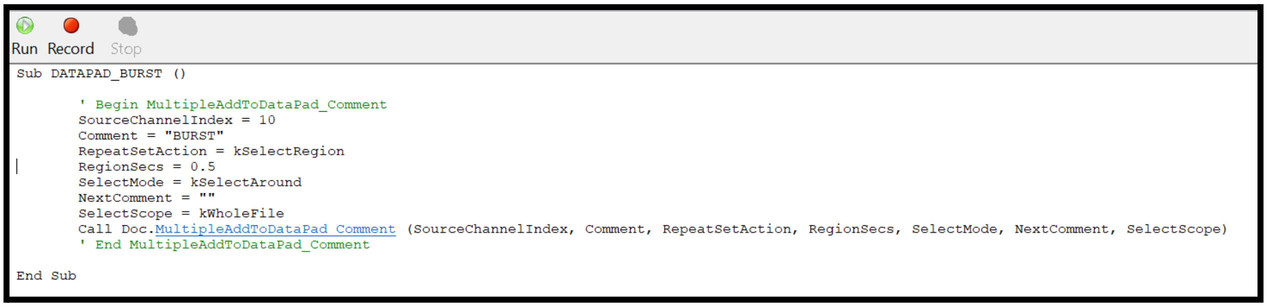
Macro → Manage → New → Type: Simple Macro, Name: DATAPAD_BURST → Check Show in toolbar → Click OK → Underneath the line Sub DATAPAD_BURST (), copy the code below:

```
SourceChannelIndex = X
Comment = "BURST"
RepeatSetAction = kSelectRegion
RegionSecs = 0.5
SelectMode = kSelectAround
NextComment = ""
SelectScope = kWholeFile
Call Doc.MultipleAddToDataPad_Comment (SourceChannelIndex, Comment, RepeatSetAction, RegionSecs, SelectMode, NextComment, SelectScope)
```

******IMPORTANT: For RegionSecs = 0.5, you may change this search window depending on the participant and/or laboratory measurement standards.***

******For the line of code highlighted blue, when copied over from the PDF into LabChart, it will sometimes break the code into two or more lines. Please make sure the line of code remains on a single line (or as a single sentence without pressing the Enter key) or else it may return an error. There may be other lines of code in the following step that do the same, so if an error is returned make sure to check that lines that are meant to be together, remain together.***

For X, use the n-1 rule to enter in as the Channel # for the Integrated/RMS MSNA Channel. For example, if the Integrated/RMS MSNA Channel is Channel 11, then enter the number 10, and the complete code should read:



```
Sub DATAPAD_BURST ()

        ' Begin MultipleAddToDataPad_Comment
        SourceChannelIndex = 10
        Comment = "BURST"
        RepeatSetAction = kSelectRegion
        RegionSecs = 0.5
        SelectMode = kSelectAround
        NextComment = ""
        SelectScope = kWholeFile
        Call Doc.MultipleAddToDataPad_Comment (SourceChannelIndex, Comment, RepeatSetAction, RegionSecs, SelectMode, NextComment, SelectScope)
        ' End MultipleAddToDataPad_Comment

End Sub
```

X. Create a Macro automating BEAT comments in the ECG Channel

This step requires the creation of several macros: (1) ECG_MAKESEL & (2) MARK_ECG

This step will create the **ECG_MAKESEL** Macro

Macro → Manage → New → Type: Simple Macro, Name: ECG_MAKESEL→ Show in toolbar → Click OK → Underneath the line Sub ECG_MAKESEL (), copy the code below:

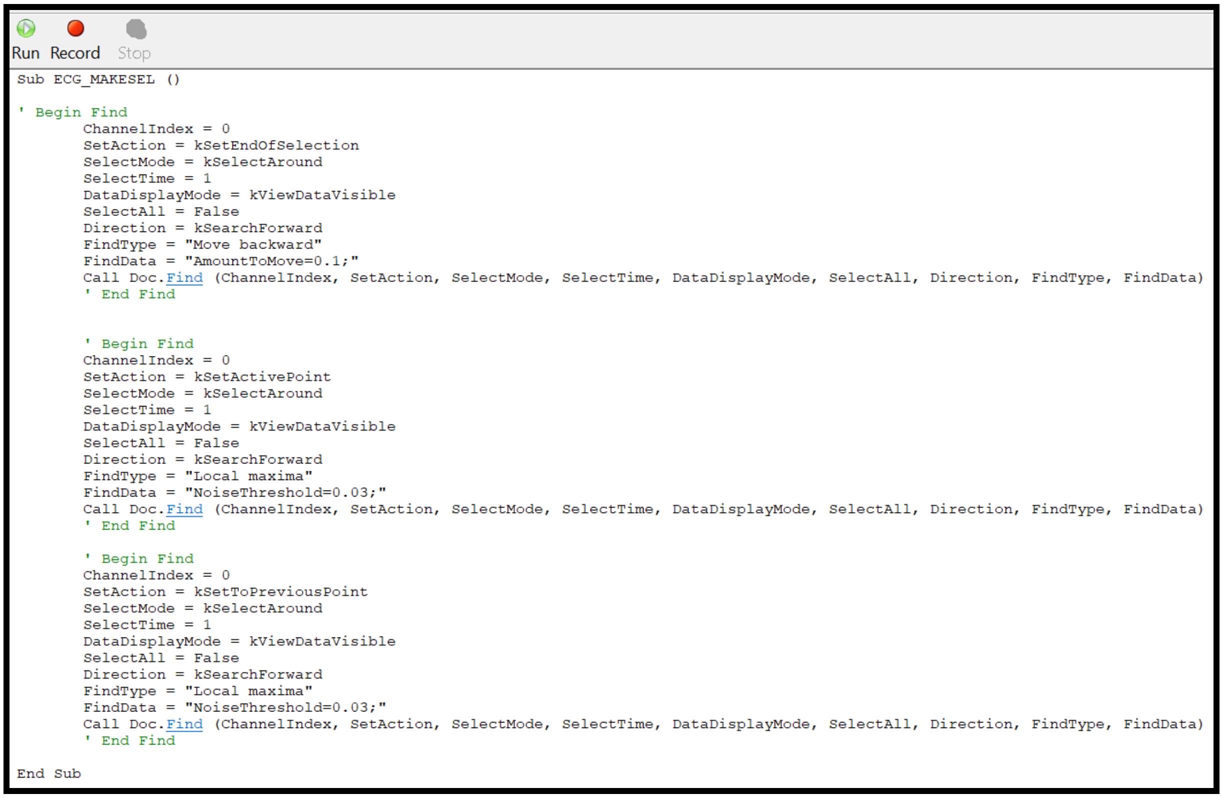
```
ChannelIndex = X
SetAction = kSetEndOfSelection
SelectMode = kSelectAround
SelectTime = 1
DataDisplayMode = kViewDataVisible
SelectAll = False
Direction = kSearchForward
FindType = "Move backward"
FindData = "AmountToMove=0.1;"
Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)

ChannelIndex =  X
SetAction = kSetActivePoint
SelectMode = kSelectAround
SelectTime = 1
DataDisplayMode = kViewDataVisible
SelectAll = False
Direction = kSearchForward
FindType = "Local maxima"
FindData = "NoiseThreshold=0.05;"
Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)

ChannelIndex =  X
SetAction = kSetToPreviousPoint
SelectMode = kSelectAround
SelectTime = 1
DataDisplayMode = kViewDataVisible
SelectAll = False
Direction = kSearchForward
FindType = "Local maxima"
FindData = "NoiseThreshold=0.05;"
Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)
```

The NoiseThreshold is set to 0.05 in this example. The noise threshold means that

Depending on the quality and/or presence of artifacts in the ECG signal, the threshold may need to be adjusted. More detail is provided in a subsequent section, "Performing Burst Identification and Analysis."

******For the lines of code highlighted blue, when copied over from the PDF into LabChart, it will sometimes break the code into two or more lines. Please make sure the line of code remains on a single line (or as a single sentence without pressing the Enter key) or else it may return an error. There may be other lines of code in the following step that do the same, so if an error is returned make sure to check that lines that are meant to be together, remain together.***

For X, use the n-1 rule to enter in as the Channel # for the ECG Channel. For example, if the ECG Channel is Channel 1, then enter the number 0, and the complete code should read:



```
Sub ECG_MAKESEL ()

' Begin Find
        ChannelIndex = 0
        SetAction = kSetEndOfSelection
        SelectMode = kSelectAround
        SelectTime = 1
        DataDisplayMode = kViewDataVisible
        SelectAll = False
        Direction = kSearchForward
        FindType = "Move backward"
        FindData = "AmountToMove=0.1;"
        Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)
        ' End Find


        ' Begin Find
        ChannelIndex = 0
        SetAction = kSetActivePoint
        SelectMode = kSelectAround
        SelectTime = 1
        DataDisplayMode = kViewDataVisible
        SelectAll = False
        Direction = kSearchForward
        FindType = "Local maxima"
        FindData = "NoiseThreshold=0.03;"
        Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)
        ' End Find

        ' Begin Find
        ChannelIndex = 0
        SetAction = kSetToPreviousPoint
        SelectMode = kSelectAround
        SelectTime = 1
        DataDisplayMode = kViewDataVisible
        SelectAll = False
        Direction = kSearchForward
        FindType = "Local maxima"
        FindData = "NoiseThreshold=0.03;"
        Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)
        ' End Find

End Sub
```

XI. This step will create the next macro needed to mark BEAT comments in the ECG channel: MARK_ECG.

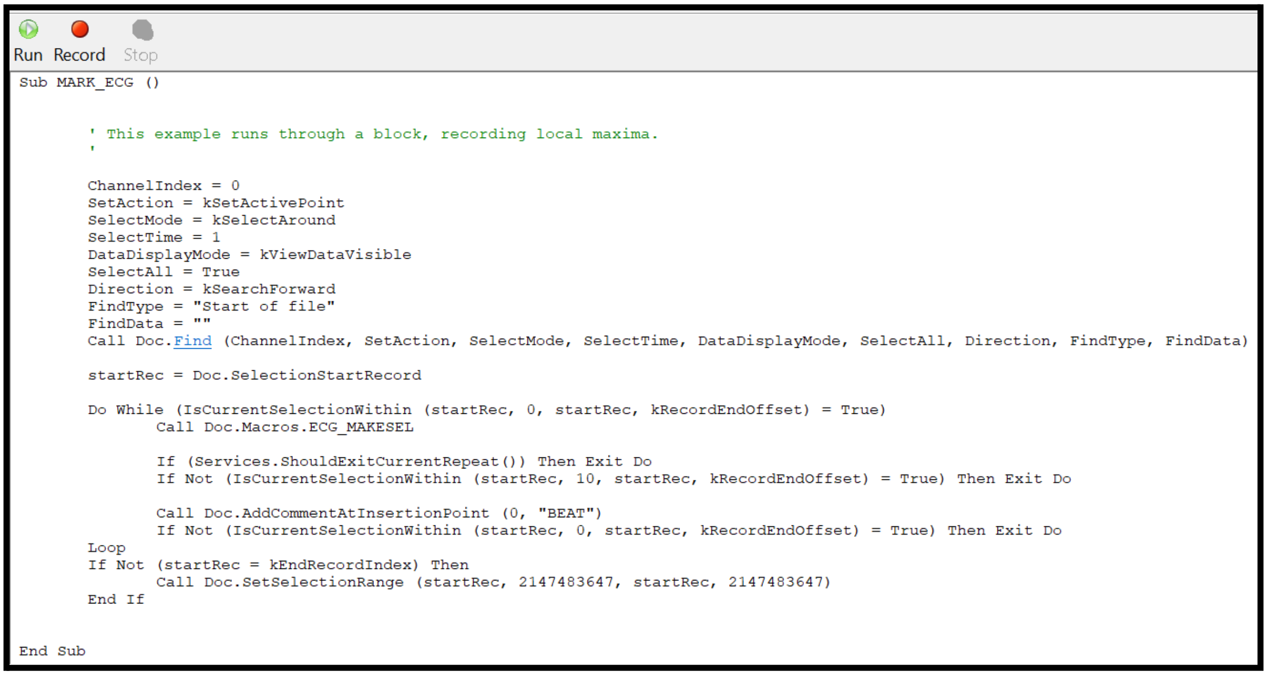
Macro → Manage → New → Type: Simple Macro, Name: MARK_ECG → Show in toolbar → Click OK → Underneath the line Sub MARK_ECG (), copy the code below:

```
ChannelIndex = X
SetAction = kSetActivePoint
SelectMode = kSelectAround
SelectTime = 1
DataDisplayMode = kViewDataVisible
SelectAll = True
Direction = kSearchForward
FindType = "Start of file"
FindData = ""
Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)
startRec = Doc.SelectionStartRecord
Do While (IsCurrentSelectionWithin (startRec, 0, startRec, kRecordEndOffset) = True)
Call Doc.Macros.ECG_MAKESEL
If (Services.ShouldExitCurrentRepeat()) Then Exit Do
If Not (IsCurrentSelectionWithin (startRec, 10, startRec, kRecordEndOffset) = True) Then Exit Do
Call Doc.AddCommentAtInsertionPoint (X, "BEAT")
If Not (IsCurrentSelectionWithin (startRec, 0, startRec, kRecordEndOffset) = True) Then Exit Do
Loop
If Not (startRec = kEndRecordIndex) Then
Call Doc.SetSelectionRange (startRec, 2147483647, startRec, 2147483647)
End If

End Sub
```

*This macro "calls" the **ECG_MAKESEL** macro, and essentially uses this macro to find local maxima (or each R-wave in the ECG tracing), and then places a BEAT comment between each R-wave.*

For X, use the n-1 rule to enter in as the Channel # for the ECG Channel. For example, if the ECG Channel is Channel

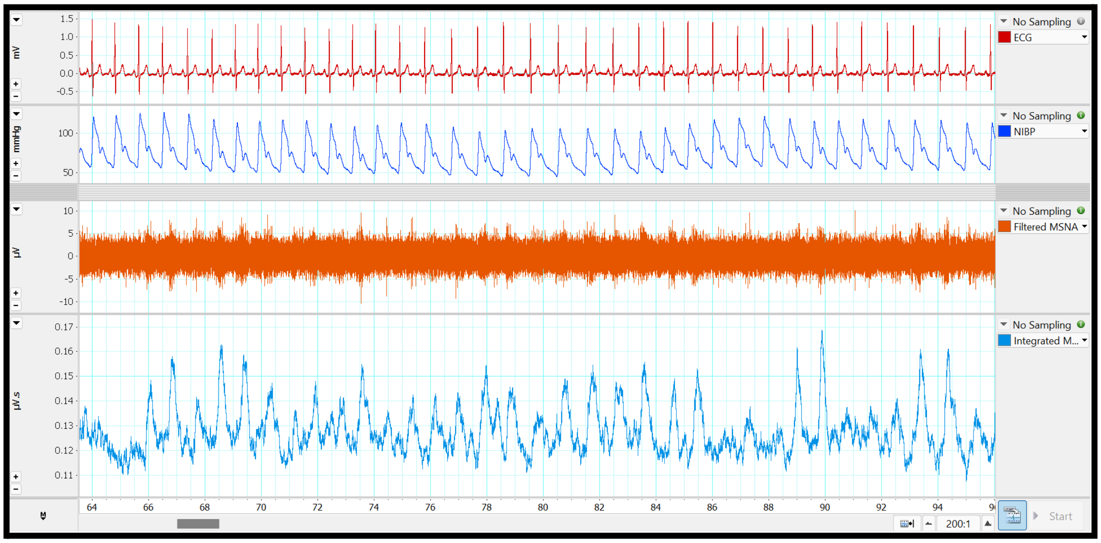
1, then enter the number 0, and the complete code should read:


Run
Record
Stop

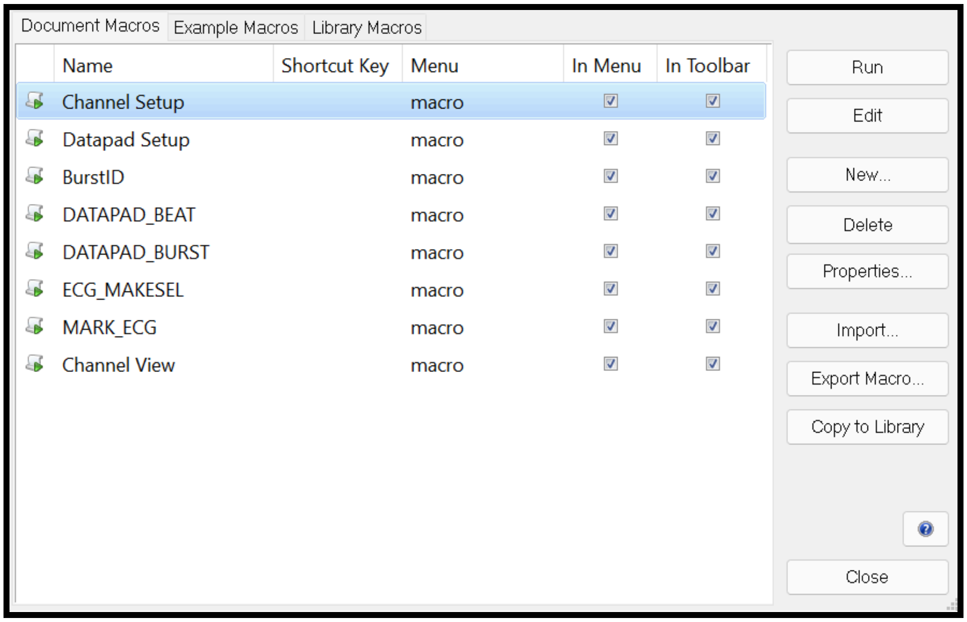
```
Sub MARK_ECG ()

        ' This example runs through a block, recording local maxima.
        '

        ChannelIndex = 0
        SetAction = kSetActivePoint
        SelectMode = kSelectAround
        SelectTime = 1
        DataDisplayMode = kViewDataVisible
        SelectAll = True
        Direction = kSearchForward
        FindType = "Start of file"
        FindData = ""
        Call Doc.Find (ChannelIndex, SetAction, SelectMode, SelectTime, DataDisplayMode, SelectAll, Direction, FindType, FindData)

        startRec = Doc.SelectionStartRecord

        Do While (IsCurrentSelectionWithin (startRec, 0, startRec, kRecordEndOffset) = True)
                Call Doc.Macros.ECG_MAKESEL

                If (Services.ShouldExitCurrentRepeat()) Then Exit Do
                If Not (IsCurrentSelectionWithin (startRec, 10, startRec, kRecordEndOffset) = True) Then Exit Do

                Call Doc.AddCommentAtInsertionPoint (0, "BEAT")
                If Not (IsCurrentSelectionWithin (startRec, 0, startRec, kRecordEndOffset) = True) Then Exit Do
        Loop
        If Not (startRec = kEndRecordIndex) Then
                Call Doc.SetSelectionRange (startRec, 2147483647, startRec, 2147483647)
        End If


End Sub
```

XII. **Record a macro to automate setting up the channel view.** The key channels are the ECG, NIBP, raw or filtered MSNA (depending on preference), and the integrated/RMS MSNA channel.

Macro → Manage → New → Type: Simple Macro, Name: Channel View → Show in toolbar → Click OK → Record setting up the channel view, similar to the image below:


mV
1.5
1.0
0.5
0.0
-0.5
No Sampling
ECG
mmHg
100
50
No Sampling
NIBP
µV
10
5
0
-5
-10
No Sampling
Filtered MSNA
µV.s
0.17
0.16
0.15
0.14
0.13
0.12
0.11
No Sampling
Integrated M...
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
200:1
Start

The setup macro list should look similar to this:



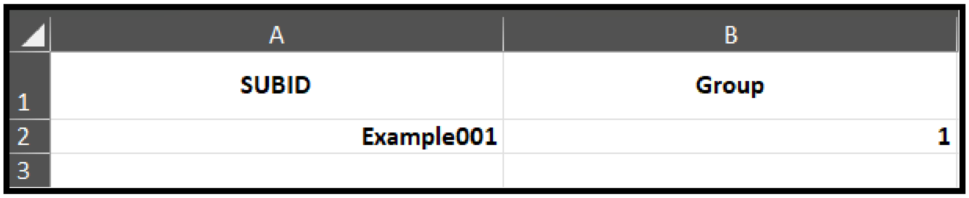
Name	Shortcut Key	Menu	In Menu	In Toolbar
Channel Setup		macro	☑	☑
Datapad Setup		macro	☑	☑
BurstID		macro	☑	☑
DATAPAD_BEAT		macro	☑	☑
DATAPAD_BURST		macro	☑	☑
ECG_MAKESEL		macro	☑	☑
MARK_ECG		macro	☑	☑
Channel View		macro	☑	☑

XIII. **TIP!:** Combine macros to expedite the setup process. Create a new macro, *run* the macros that will be combined, save as a parent macro.

Setting up Excel File for Analysis

I. Open the BurstID_TemplateFile on Excel. Make sure Excel is allowed to run macros. A popup in Excel may occur if macros need to be enabled for this file.

II. In the CumulativeSheet tab, enter the specified participant's SUBID and Group # (if applicable; for example, females are Group 1 and males are Group 2):



	A	B
1	**SUBID**	**Group**
2	**Example001**	**1**
3		

III. Excel Workbook Tabs:

CumulativeSheet: all data will automatically populate into here from subsequent sheets

LabChart_CorrectedInput_Rest: this sheet will calculate resting MSNA variables

LabChartInput_BURST_Rest: data here will be pasted from the Data Pad after extraction based on BURST comments from resting LabChart files

LabChartInput_BEAT_Rest: data here will be pasted from the Data Pad after extraction based on BEAT comments from resting LabChart files

MSNA Outcomes_Rest: all resting MSNA outcomes will be populated here for ease of viewing

BurstTransduction_MAP: resting MAP transduction data will be populated here

NonBurstTransduction_MAP: resting MAP nonburst transduction data will be populated here

BurstTransduction_DBP: resting DBP transduction data will be populated here

NonBurstTransduction_DBP: resting DBP nonburst transduction data will be populated here

CPT/Other Stressor Tabs: While these tabs are set up to analyze a stressor with a 3-min baseline, 2-min stressor, and 5-min recovery, these tabs are modifiable to accommodate different durations of analyses. If clarity is needed on how to adjust formulas and outcome variable nomenclature, please contact the corresponding author.

MSNA Outcomes_CPT: all MSNA outcomes from CPT or other stressors will be populated here for ease of viewing. Modify this sheet as needed to analyze various stressors (Stroop Color and Word Test, mental arithmetic, etc.) that contain a baseline, stressor, and recovery.

LabChart_CorrectedInput_CPT: this sheet will calculate MSNA variables for CPT/other stressors

LabChartInput_BURST_CPT: data here will extracted based on BURST comments from CPT/other stressor LabChart files

LabChartInput_BEAT_CPT: data here will extracted based on BEAT comments from CPT/other stressor LabChart files

EPR (Exercise Pressor Reflex) Stressor Tabs:
While these tabs are set up to analyze exercise pressor reflex testing (3-min baseline, 2-min handgrip [IHG], 3-min post-exercise circulatory occlusion [PECO]), this tab is modifiable to accommodate different durations of analyses. For example, the IHG outcomes are calculated as 25% of the IHG duration, and can thus accommodate a wide range of exercise times. However, PECO is calculated in 30-ss epochs, but can accommodate a wide range of durations. If clarity is needed on how to adjust formulas and outcome variable nomenclature, please contact the corresponding author.

MSNA Outcomes_EPR: all exercise pressor reflex testing MSNA outcomes will be populated here for ease of viewing. This sheet can be used for any stressor with four segmented parts. Modify this sheet as needed.

LabChart_CorrectedInput_EPR: this sheet will calculate EPR/other stressor MSNA variables

LabChartInput_BURST_EPR: data here will be pasted from the Data Pad after extraction based on BURST comments from EPR/other stressor LabChart files

LabChartInput_BEAT_EPR: data here will be pasted from the Data Pad after extraction based on BEAT comments from EPR/other stressor LabChart files

IV. Move onto the next section.

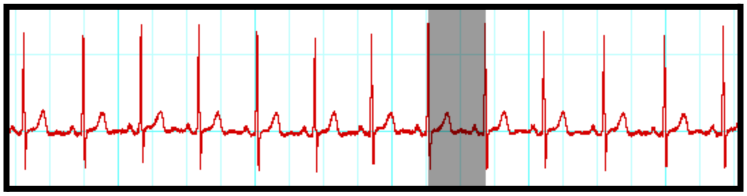
Performing Burst Identification and Analysis

I. **IMPORTANT NOTE:** Save the current file with all the created macros so that they can be reuploaded into all subsequent files. In new files choose: ***Macro → Manage → Import → Navigate to saved macro file and import into new file.***

II. Run the **Channel Setup** macro.

III. Run the **Datapad Setup** macro.

IV. Click anywhere on the ECG tracing. Run the **ECG_MAKESEL** macro a few times. It should highlight from each R-R interval on the ECG tracing as seen below:



If this macro is skipping beats, or erroneously selecting other components of the tracing that are not the R-wave, stop the macro, delete all of the BEAT comments, and troubleshoot by editing certain lines of the code:

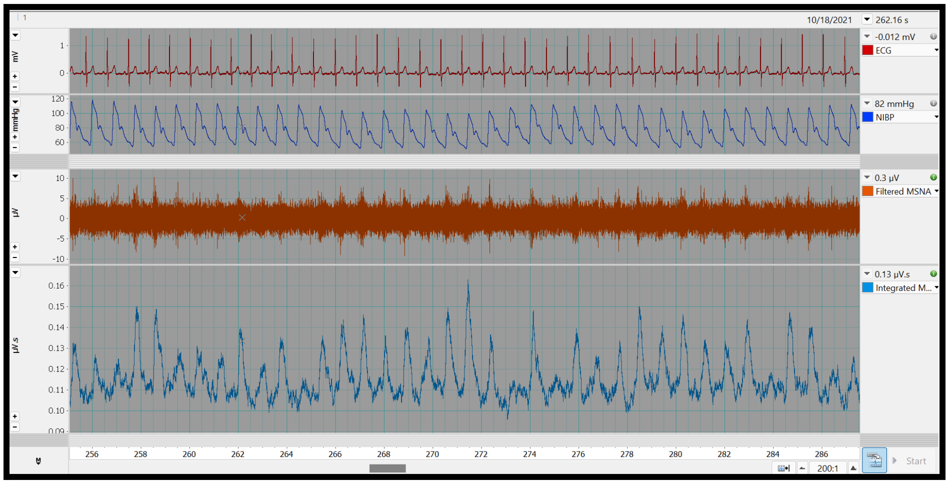
Option 1: Either increase or decrease the noise threshold. Be strategic, and notice what increasing and decreasing this number does to detecting components of the tracing. Remember, there are two places in the ECG_MAKESEL macro where this number will need to be changed.

Option 2: Change the macro to detect the local minimas rather than local maximas.

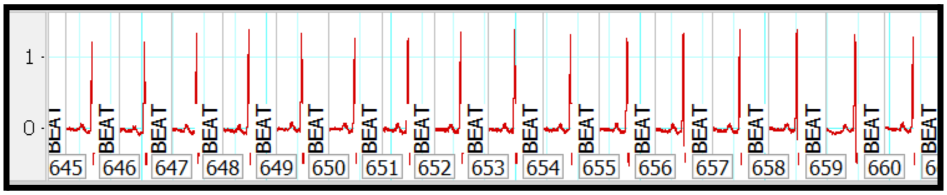
FindType = "Local maxima" change to FindType = "Local minima"

If neither of these options work for all beats, it may be necessary to run the MARK_ECG macro, then manually enter some BEAT comments or to move the placement of some BEAT comments. Once all R-R intervals have been marked by a BEAT comment, proceed to the next step. During data collection, ensuring that there is a clean ECG signal will reduce the likelihood of having to manually enter and/or edit BEAT comments placed by the macro.

V. Double-click on the timestamp scale at the bottom of the screen to highlight the entire file:

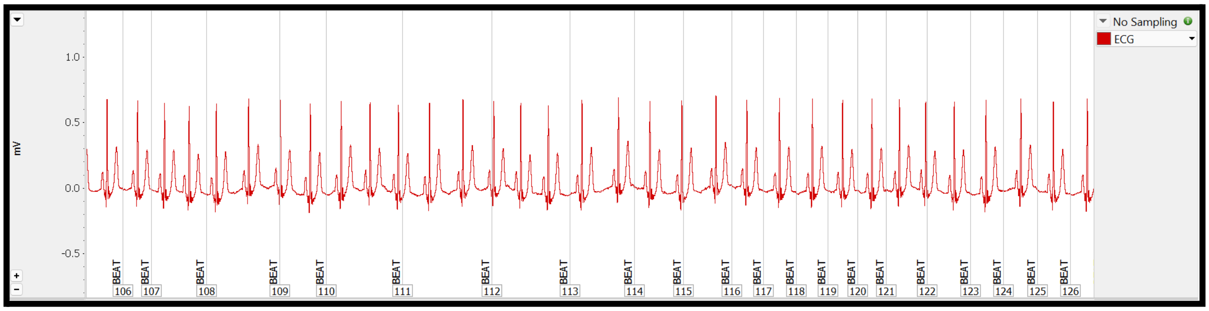

10/18/2021
262.16 s
-0.012 mV
ECG
82 mmHg
NIBP
0.3 µV
Filtered MSNA
0.13 µV.s
Integrated M...
mV
mmHg
µV
µV.s
256
258
260
262
264
266
268
270
272
274
276
278
280
282
284
286
200:1
Start

VI. Run the **MARK_ECG** Macro. This macro will add a BEAT comment between each R-wave. Example of a clean ECG file below:

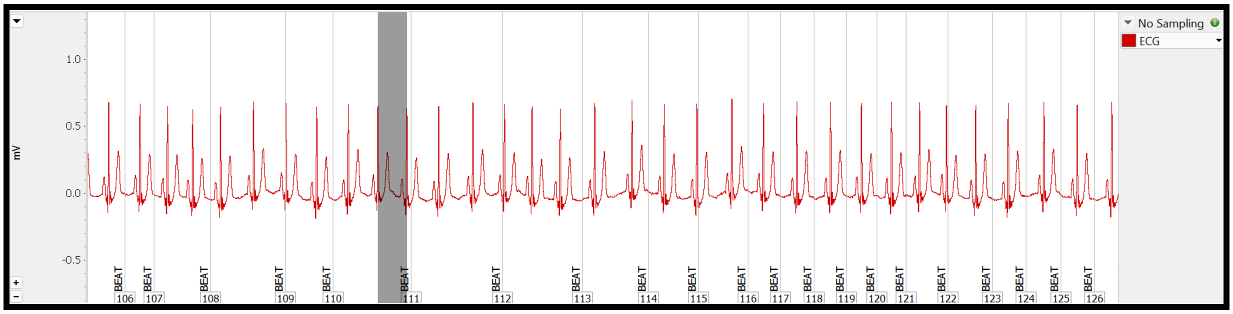

1
0
BEAT
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660

—-

***As stated in **Section X. Create a Macro automating BEAT comments in the ECG Channel,** it is possible you will need to adjust the noise threshold in the ECG_MAKESEL Macro. The image below is an example of how movement artifacts or dry electrodes may impact the ECG tracing. By definition, the noise threshold is the % of the range of the channel that will be ignored when searching for local maxima. So, if the noise threshold is set to 0.05 and the range of your ECG channel is 100mV, then anything below 5mV will be ignored when searching for local maxima.

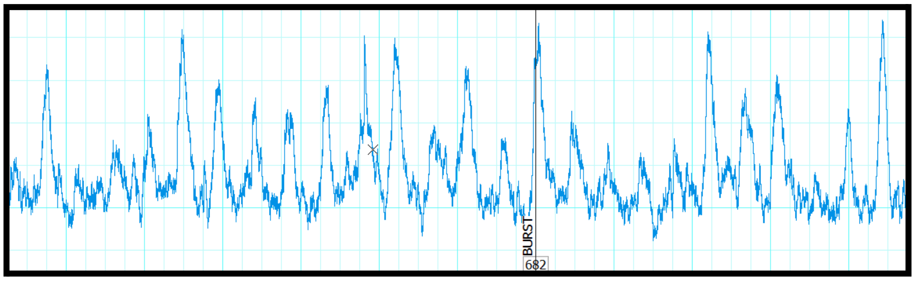

No Sampling
ECG
mV
1.0
0.5
0.0
-0.5
BEAT
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126

As you can see in the image above, there are missing ‘BEAT’ comments.


No Sampling
ECG
mV
1.0
0.5
0.0
-0.5
BEAT 106
BEAT 107
BEAT 108
BEAT 109
BEAT 110
BEAT 111
BEAT 112
BEAT 113
BEAT 114
BEAT 115
BEAT 116
BEAT 117
BEAT 118
BEAT 119
BEAT 120
BEAT 121
BEAT 122
BEAT 123
BEAT 124
BEAT 125
BEAT 126

In the ECG_MAKESEL macro, the noise threshold was dropped from 0.05 to 0.03 and now the correct window from R-wave to R-wave is being highlighted. Before re-running the MARK_ECG macro, please delete all existing ECG comments (double click on a 'BEAT' comment, then highlight the list and press delete, do not individually delete comments!). Run the ECG_MAKESEL macro a few times to ensure no beats are being missed, then highlight the entire file and re-run the MARK_ECG macro.

VII. It is now time to perform burst identification. In the integrated MSNA/RMS MSNA, click on the peak of each burst, then run the **Manual_BurstID** macro:



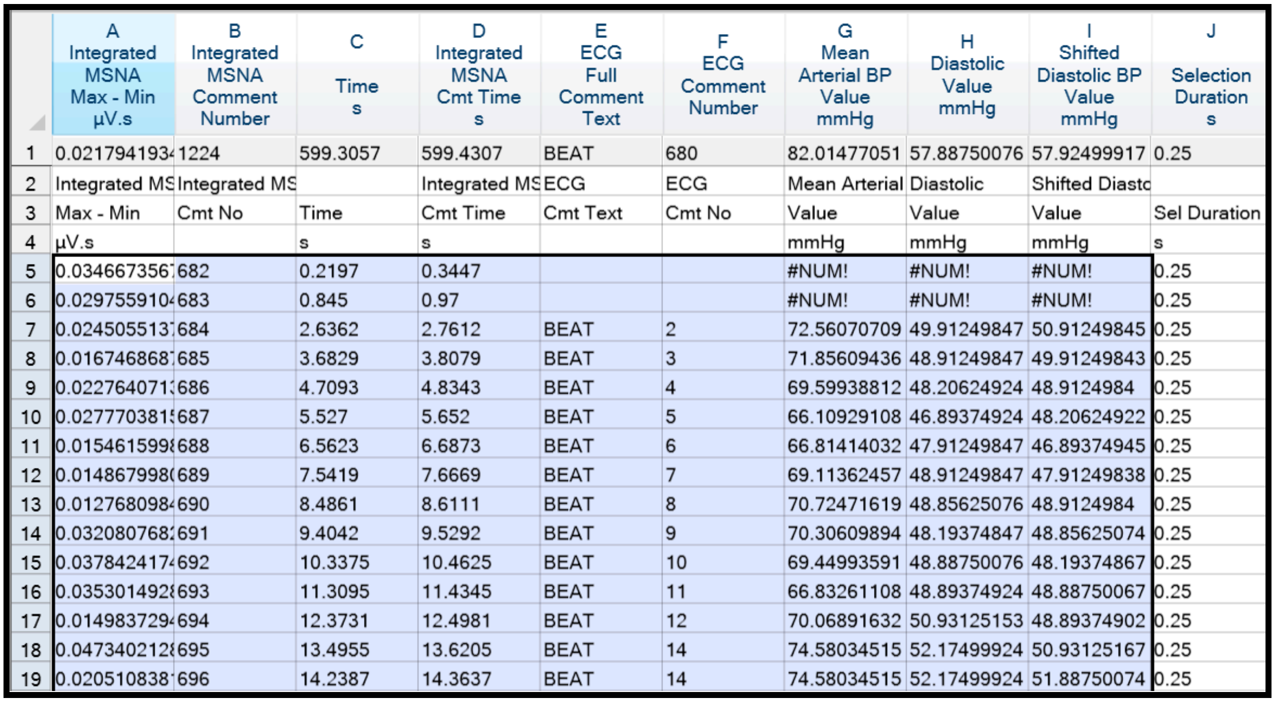
This will add a BURST comment at the peak of each burst. **IMPORTANT NOTE**: (1) automated methods may be developed so that each burst does not need to be manually annotated, and (2) the duration from the preceding R-wave may be checked to ensure that bursts fall within the normal range of burst propagation. The numerical order of the BURST comments will not affect subsequent data analysis, so comments do not need to be shifted if an additional BURST comment is added between two existing BURST comments. Once burst marking for the entire file is completed, move onto the next step.

VIII. Open the **BurstID_TemplateFile_Date*** Excel workbook. Allow macros to be run in the Excel Workbook after downloading and opening the file. There should be no security risks to do this.

IX. Go to the **LabChartInput_BURST_Rest** tab.

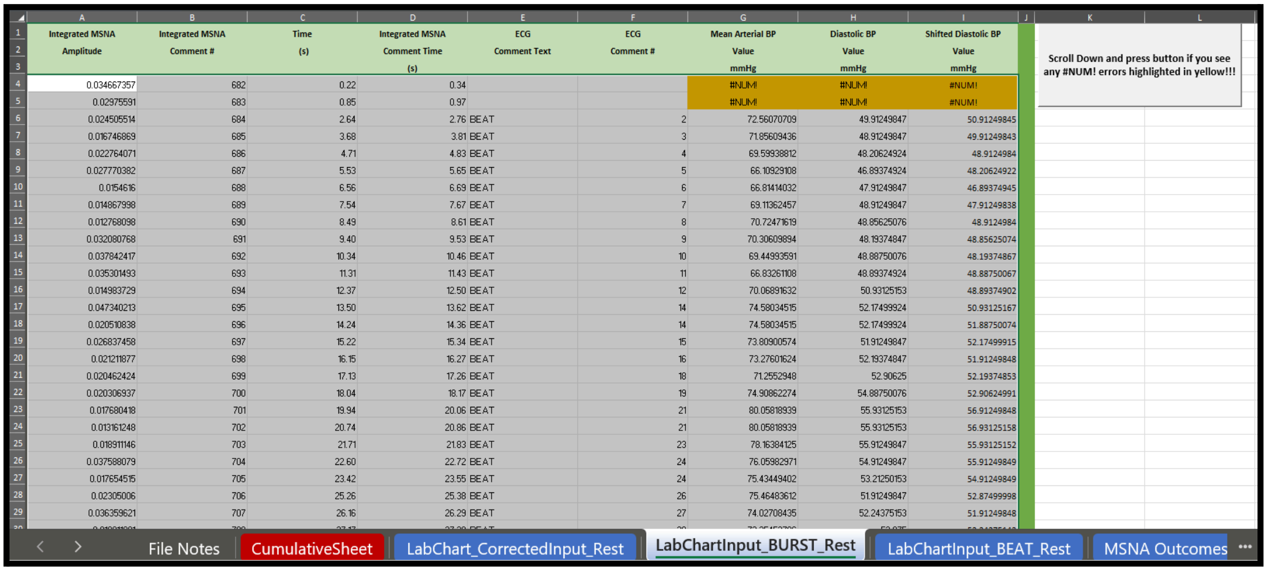
X. Run the **DATAPAD_BURST** macro. This macro will look for BURST comments and add all pertinent, associated data into the Data Pad. Using the "Cell edit mode," copy columns A-I of the Data Pad (not the column names) into the **LabChartInput_BURST_Rest** tab:

Step 1: Add all data into Data Pad from BURST comments by running the DATAPAD_BURST macro.



	A Integrated MSNA Max - Min µV.s	B Integrated MSNA Comment Number	C Time s	D Integrated MSNA Cmt Time s	E ECG Full Comment Text	F ECG Comment Number	G Mean Arterial BP Value mmHg	H Diastolic Value mmHg	I Shifted Diastolic BP Value mmHg	J Selection Duration s
1	0.021794193	1224	599.3057	599.4307	BEAT	680	82.01477051	57.88750076	57.92499917	0.25
2	Integrated MS	Integrated MS		Integrated MS	ECG	ECG	Mean Arterial	Diastolic	Shifted Diastc	
3	Max - Min	Cmt No	Time	Cmt Time	Cmt Text	Cmt No	Value	Value	Value	Sel Duration
4	µV.s		s	s			mmHg	mmHg	mmHg	s
5	0.034667356	682	0.2197	0.3447			#NUM!	#NUM!	#NUM!	0.25
6	0.029755910	683	0.845	0.97			#NUM!	#NUM!	#NUM!	0.25
7	0.024505513	684	2.6362	2.7612	BEAT	2	72.56070709	49.91249847	50.91249845	0.25
8	0.016746868	685	3.6829	3.8079	BEAT	3	71.85609436	48.91249847	49.91249843	0.25
9	0.022764071	686	4.7093	4.8343	BEAT	4	69.59938812	48.20624924	48.9124984	0.25
10	0.027770381	687	5.527	5.652	BEAT	5	66.10929108	46.89374924	48.20624922	0.25
11	0.015461599	688	6.5623	6.6873	BEAT	6	66.81414032	47.91249847	46.89374945	0.25
12	0.014867998	689	7.5419	7.6669	BEAT	7	69.11362457	48.91249847	47.91249838	0.25
13	0.012768098	690	8.4861	8.6111	BEAT	8	70.72471619	48.85625076	48.9124984	0.25
14	0.032080768	691	9.4042	9.5292	BEAT	9	70.30609894	48.19374847	48.85625074	0.25
15	0.037842417	692	10.3375	10.4625	BEAT	10	69.44993591	48.88750076	48.19374867	0.25
16	0.035301492	693	11.3095	11.4345	BEAT	11	66.83261108	48.89374924	48.88750067	0.25
17	0.014983729	694	12.3731	12.4981	BEAT	12	70.06891632	50.93125153	48.89374902	0.25
18	0.047340212	695	13.4955	13.6205	BEAT	14	74.58034515	52.17499924	50.93125167	0.25
19	0.020510838	696	14.2387	14.3637	BEAT	14	74.58034515	52.17499924	51.88750074	0.25

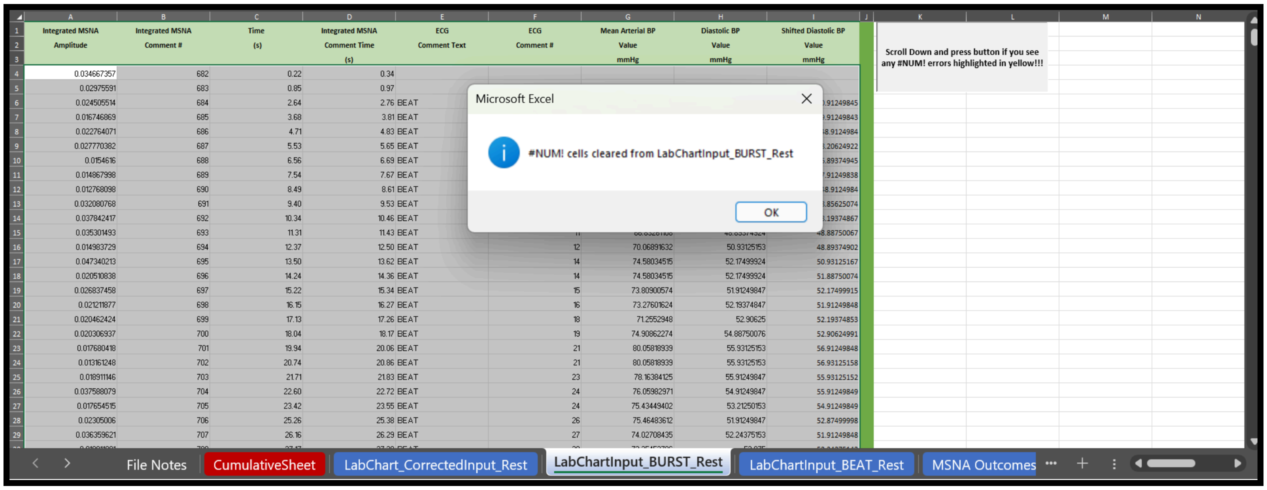
Step 2: Copy data AS SHOWN ABOVE into the LabCharInput_BURST_Rest tab



	A	B	C	D	E	F	G	H	I
1	Integrated MSNA	Integrated MSNA	Time	Integrated MSNA	ECG	ECG	Mean Arterial BP	Diastolic BP	Shifted Diastolic BP
2	Amplitude	Comment #	(s)	Comment Time	Comment Text	Comment #	Value	Value	Value
3				(s)			mmHg	mmHg	mmHg
4	0.034667357	682	0.22	0.34			#NUM!	#NUM!	#NUM!
5	0.02975591	683	0.85	0.97			#NUM!	#NUM!	#NUM!
6	0.024505514	684	2.64	2.76	BEAT	2	72.56070709	49.91249847	50.91249845
7	0.016746869	685	3.68	3.81	BEAT	3	71.85609436	48.91249847	49.91249843
8	0.022764071	686	4.71	4.83	BEAT	4	69.59938812	48.20624924	48.9124984
9	0.027770382	687	5.53	5.65	BEAT	5	66.10929108	46.89374924	48.20624922
10	0.0154616	688	6.56	6.69	BEAT	6	66.81414032	47.91249847	46.89374945
11	0.014867998	689	7.54	7.67	BEAT	7	69.11362457	48.91249847	47.91249838
12	0.012768098	690	8.49	8.61	BEAT	8	70.72471619	48.85625076	48.9124984
13	0.032080768	691	9.40	9.53	BEAT	9	70.30609894	48.19374847	48.85625074
14	0.037842417	692	10.34	10.46	BEAT	10	69.44993591	48.88750076	48.19374867
15	0.035301493	693	11.31	11.43	BEAT	11	66.83261108	48.89374924	48.88750067
16	0.014983729	694	12.37	12.50	BEAT	12	70.06891632	50.93125153	48.89374902
17	0.047340213	695	13.50	13.62	BEAT	14	74.58034515	52.17499924	50.93125167
18	0.020510838	696	14.24	14.36	BEAT	14	74.58034515	52.17499924	51.88750074
19	0.026837458	697	15.22	15.34	BEAT	15	73.80900574	51.91249847	52.17499915
20	0.021211877	698	16.15	16.27	BEAT	16	73.27601624	52.19374847	51.91249848
21	0.020462424	699	17.13	17.26	BEAT	18	71.2552948	52.90625	52.19374853
22	0.020306937	700	18.04	18.17	BEAT	19	74.90862274	54.88750076	52.90624991
23	0.017680418	701	19.94	20.06	BEAT	21	80.05818939	55.93125153	56.91249848
24	0.013161248	702	20.74	20.86	BEAT	21	80.05818939	55.93125153	56.93125158
25	0.018911146	703	21.71	21.83	BEAT	23	78.16384125	55.91249847	55.93125152
26	0.037588079	704	22.60	22.72	BEAT	24	76.05982971	54.91249847	55.91249849
27	0.017654515	705	23.42	23.55	BEAT	24	75.43449402	53.21250153	54.91249849
28	0.02305006	706	25.26	25.38	BEAT	26	75.46483612	51.91249847	52.87499998
29	0.036359621	707	26.16	26.29	BEAT	27	74.02708435	52.24375153	51.91249848

Scroll Down and press button if you see any #NUM! errors highlighted in yellow!!!

Step 3: There may be **#NUM** cells that are highlighted in yellow. This occurs due to the way LabChart clips exported data at the beginning and end of files. Press the button to the right to clear all of these cells.



Integrated MSNA Amplitude	Integrated MSNA Comment #	Time (s)	Integrated MSNA Comment Time (s)	ECG Comment Text	ECG Comment #	Mean Arterial BP Value mmHg	Diastolic BP Value mmHg	Shifted Diastolic BP Value mmHg
0.034667357	682	0.22	0.34					
0.02975591	683	0.85	0.97					
0.024505514	684	2.64	2.76	BEAT				
0.016746869	685	3.68	3.81	BEAT				
0.022764071	686	4.71	4.83	BEAT				
0.027770382	687	5.53	5.65	BEAT				
0.0154616	688	6.56	6.69	BEAT				
0.014867998	689	7.54	7.67	BEAT				
0.012768098	690	8.49	8.61	BEAT				
0.032080768	691	9.40	9.53	BEAT				
0.037842417	692	10.34	10.46	BEAT				
0.035301493	693	11.31	11.43	BEAT				48.88750067
0.014983729	694	12.37	12.50	BEAT	12	70.06891632	50.93125153	48.89374902
0.047340213	695	13.50	13.62	BEAT	14	74.58034515	52.17499924	50.93125167
0.020510838	696	14.24	14.36	BEAT	14	74.58034515	52.17499924	51.88750074
0.026837458	697	15.22	15.34	BEAT	15	73.80900574	51.91249847	52.17499915
0.021211877	698	16.15	16.27	BEAT	16	73.27601624	52.19374847	51.91249848
0.020462424	699	17.13	17.26	BEAT	18	71.2552948	52.90625	52.19374853
0.020306937	700	18.04	18.17	BEAT	19	74.90862274	54.88750076	52.90624991
0.017680418	701	19.94	20.06	BEAT	21	80.05818939	55.93125153	56.91249848
0.013161248	702	20.74	20.86	BEAT	21	80.05818939	55.93125153	56.93125158
0.018911146	703	21.71	21.83	BEAT	23	78.16384125	55.91249847	55.93125152
0.037588079	704	22.60	22.72	BEAT	24	76.05982971	54.91249847	55.91249849
0.017654515	705	23.42	23.55	BEAT	24	75.43449402	53.21250153	54.91249849
0.02305006	706	25.26	25.38	BEAT	26	75.46483612	51.91249847	52.87499998
0.036359621	707	26.16	26.29	BEAT	27	74.02708435	52.24375153	51.91249848

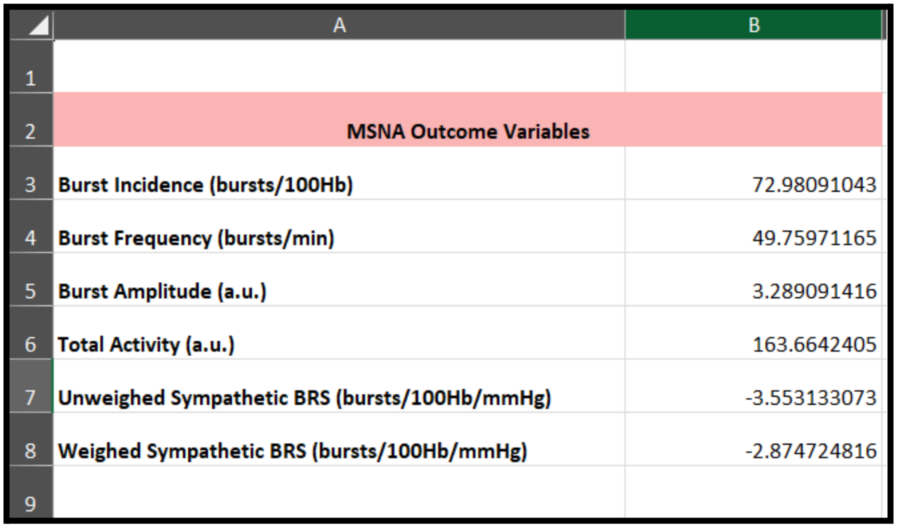
Microsoft Excel

#NUM! cells cleared from LabChartInput_BURST_Rest

Scroll Down and press button if you see any #NUM! errors highlighted in yellow!!!

XI. *COMPLETELY* clear the Data Pad in LabChart. Then, highlight the entire file, and run the **DATAPAD_BEAT** macro. This macro will look for BURST comments and add all pertinent, associated data into the Data Pad. Copy the Data Pad (not the column names) into the **LabChartInput_BEAT_Rest** tab. Press the button to eliminate all #NUM! values.

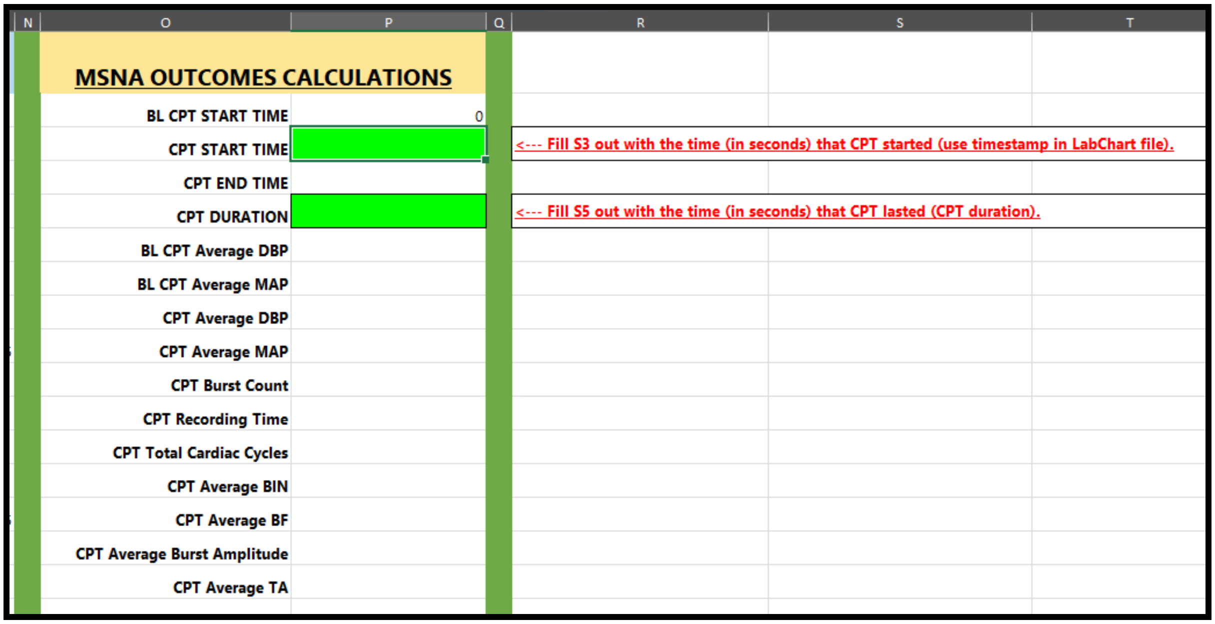
XII. Now go to the **LabChart_CorrectedInput** tab. If burst area is used instead of burst amplitude, please change the text acros this tab and the entire Excel workbook. This tab will calculate many resting MSNA outcomes. Scan the tab for any error cells and clear or modify, if applicable. Go to the **MSNA_OutcomesRest** tab. If any data are blank, then that means there may be some errors in either the way the data was exported into LabChart or in the **LabChart_CorrectedInput_Rest** sheet. Please double check that the data is being calculated as intended and that data were exported into the correct columns in the **LabChartInput_BEAT_Rest** and **LabChartInput_BURST_Rest** tabs. The Unweighted and Weighted sBRS slopes will only appear if their respective correlation coefficients are $\geq$0.7. An example of this tab is shown below:



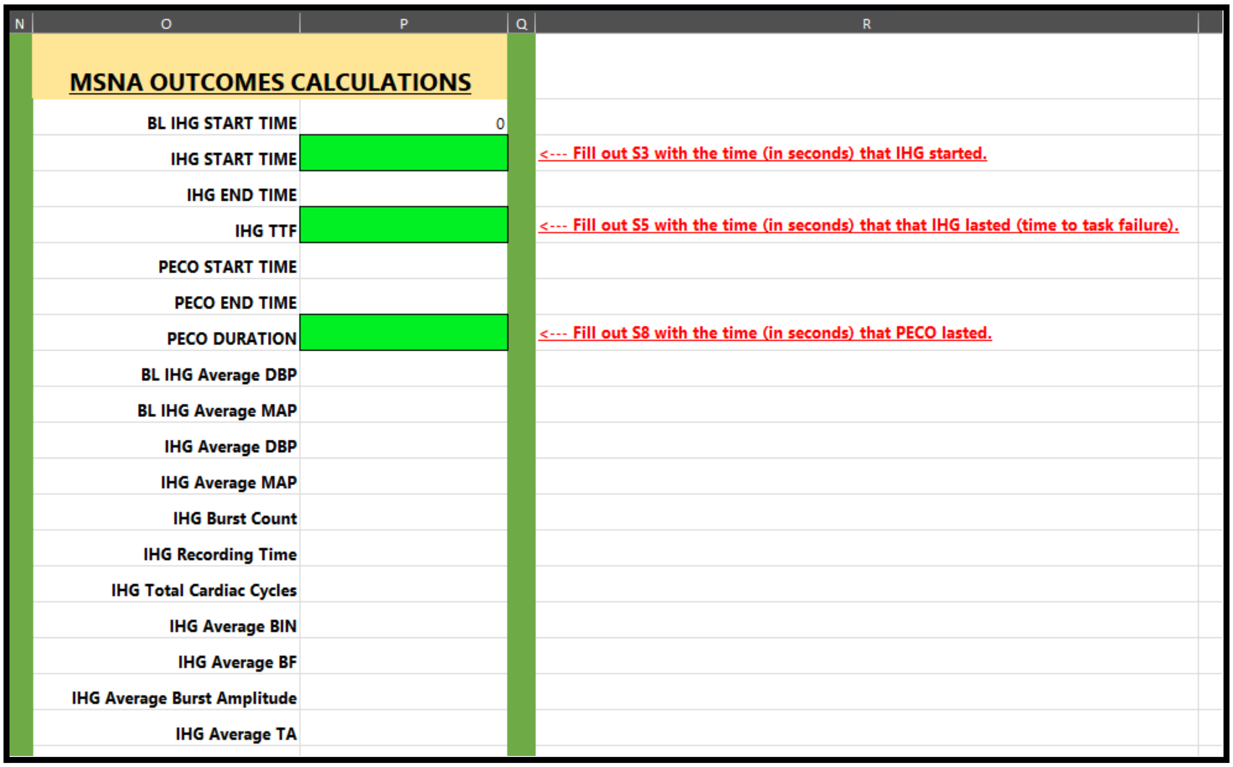
MSNA Outcome Variables	
Burst Incidence (bursts/100Hb)	72.98091043
Burst Frequency (bursts/min)	49.75971165
Burst Amplitude (a.u.)	3.289091416
Total Activity (a.u.)	163.6642405
Unweighed Sympathetic BRS (bursts/100Hb/mmHg)	-3.553133073
Weighed Sympathetic BRS (bursts/100Hb/mmHg)	-2.874724816

XIII. Go to the Transduction tabs. Check for any errors or missing data and modify, if applicable. These tabs will calculate overall transduction, overall nonburst transduction, burst amplitude quartile transduction, and BP tertile transduction. **If there were any baseline shifts or movement of the microelectrode, burst amplitude analysis may not be accurate.**.

XIV. If using the CPT or EPR (other stressor) tabs, follow the process outlined above for the specified BEAT or BURST data. Ensure, in their respective CorrectedInput tabs, to input the specified timepoints/information in the cells that are colored **green.** Use these tabs to perform analyses on a wide range of stressors. Be familiar with the structure and some of the code before making any changes.



N	O	P	Q	R	S	T
	MSNA OUTCOMES CALCULATIONS					
	BL CPT START TIME	0				
	CPT START TIME	🟩		<--- Fill S3 out with the time (in seconds) that CPT started (use timestamp in LabChart file).		
	CPT END TIME					
	CPT DURATION	🟩		<--- Fill S5 out with the time (in seconds) that CPT lasted (CPT duration).		
	BL CPT Average DBP					
	BL CPT Average MAP					
	CPT Average DBP					
	CPT Average MAP					
	CPT Burst Count					
	CPT Recording Time					
	CPT Total Cardiac Cycles					
	CPT Average BIN					
	CPT Average BF					
	CPT Average Burst Amplitude					
	CPT Average TA					



N	O	P	Q	R
	MSNA OUTCOMES CALCULATIONS			
	BL IHG START TIME	0		
	IHG START TIME	🟩		<--- Fill out S3 with the time (in seconds) that IHG started.
	IHG END TIME			
	IHG TTF	🟩		<--- Fill out S5 with the time (in seconds) that that IHG lasted (time to task failure).
	PECO START TIME			
	PECO END TIME			
	PECO DURATION	🟩		<--- Fill out S8 with the time (in seconds) that PECO lasted.
	BL IHG Average DBP			
	BL IHG Average MAP			
	IHG Average DBP			
	IHG Average MAP			
	IHG Burst Count			
	IHG Recording Time			
	IHG Total Cardiac Cycles			
	IHG Average BIN			
	IHG Average BF			
	IHG Average Burst Amplitude			
	IHG Average TA			

Fixing Errors

Follow these steps if any errors arise in completing this analysis:

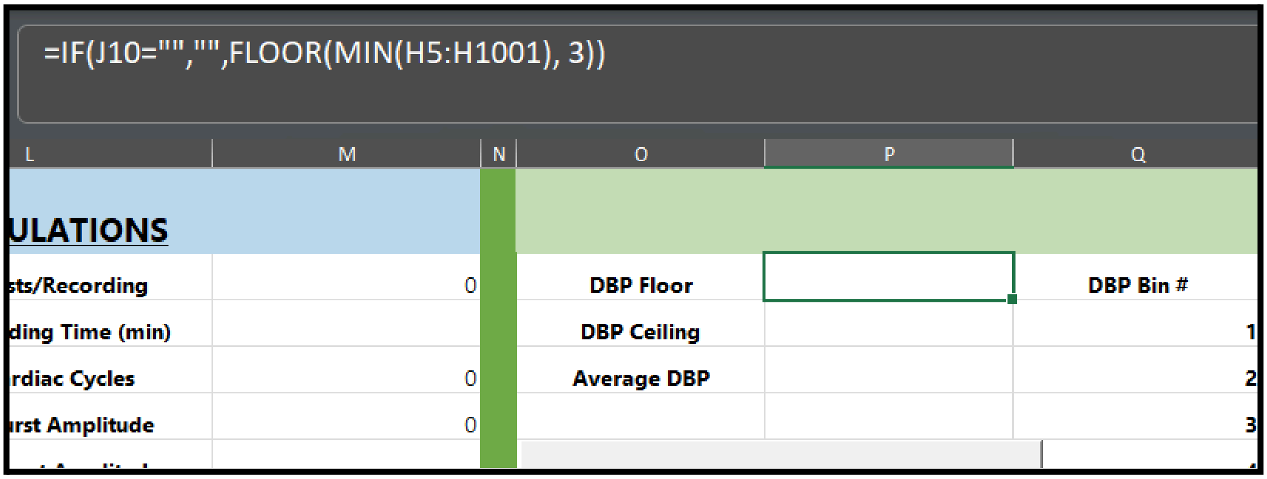
1. Ensure that data is copied appropriately from LabChart. Make sure all BEAT comments are ***only*** attached to the ECG channel and all BURST comments are ***only*** attached to the Integrated/RMS MSNA Channel. Ensure that all pre-existing comments are deleted from LabChart.

2.

2. Make sure the timestamp is changed appropriately, as previously described, to be in seconds from the start of the file.

3. Ensure that the Data Pad columns are accurately aligned with the Excel columns.

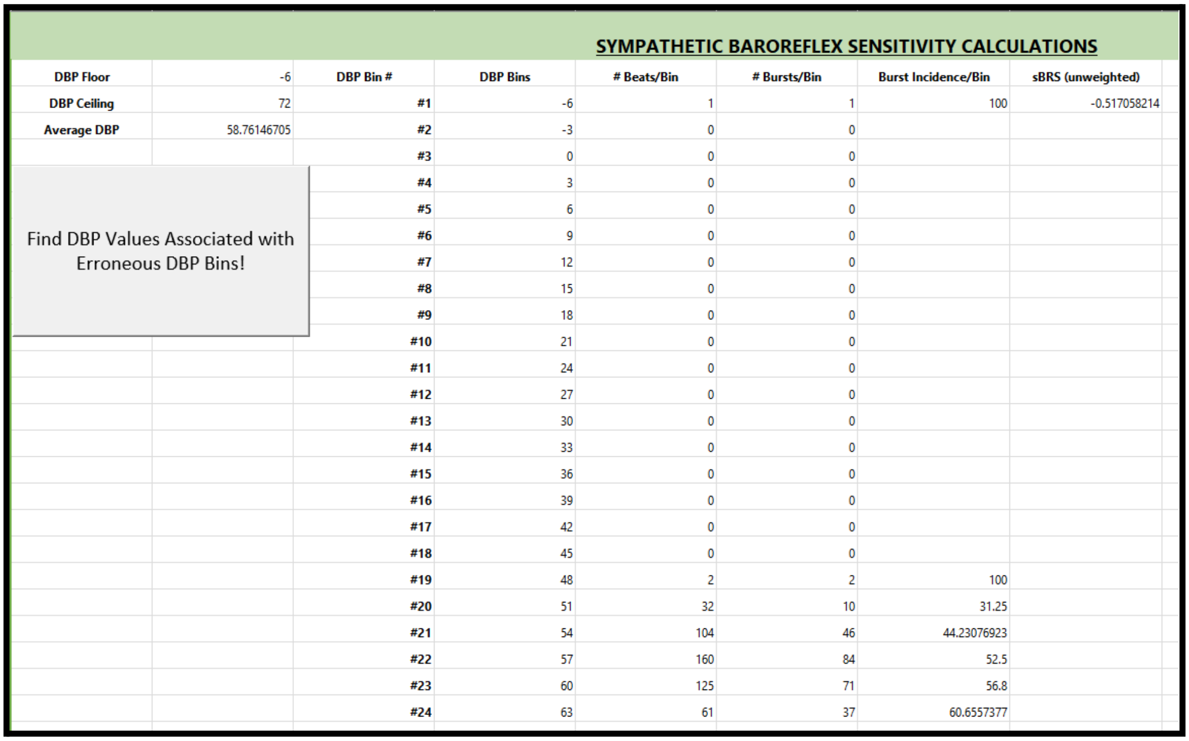
4. Some of the formulas in certain cells in the workbook are conditional. Blank cells should be checked to ensure the reference cells are not blank themselves. For example, in the image below, the data present in the DBP Floor (Cell P2) and DBP Ceiling (Cell P3) cell formulas are conditional on Cell J10 having a value. Thus, if Cell J10 is blank (depending on the dataset), this condition should either be deleted, or another cell that has a value that was exported from LabChart should be selected (i.e., J10 should be changed to the number of a different cell that has a value). In the MSNA_Outcomes tabs for rest and during stressors, many of the calculated cells to display analyzed MSNA data have an IFERROR function, so if the cells are blank it means there is an error in the cells that are being calculated.



```
=IF(J10="","",FLOOR(MIN(H5:H1001), 3))
```

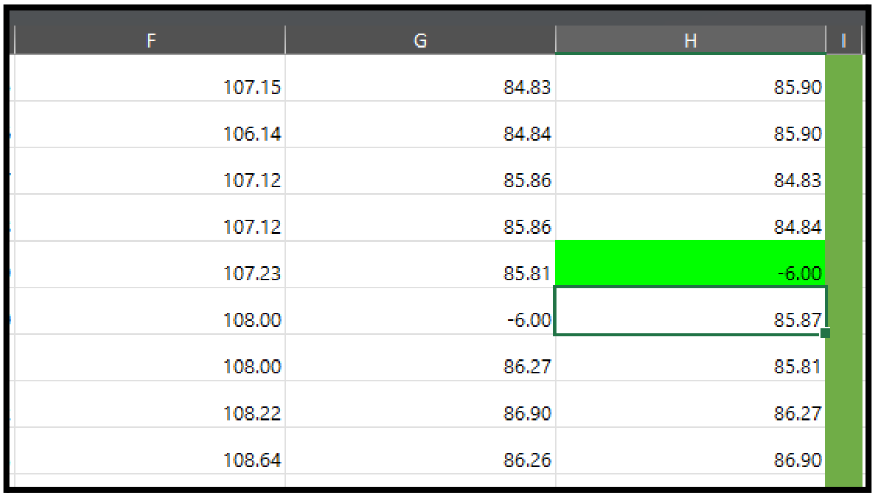
L	M	N	O	P	Q
ULATIONS					
ts/Recording	0		DBP Floor		DBP Bin #
ding Time (min)			DBP Ceiling		1
rdiac Cycles	0		Average DBP		2
rst Amplitude	0				3

5. For the sympathetic baroreflex calculation, incorrect pressure bins may be generated if there are erroneous data present in Column H, the shifted DBP column (see image below; a DBP of −6 mmHg is not possible). If such bins occur, the button titled "Find DBP Values Associated with Erroneous DBP Bins!" should be clicked and the Bin # entered. In Column H (send figure below), the DBP values associated with each bin will be highlighted. If any erroneous DBP values are noticed (such as in the image below, DBP cannot physiologically be a negative value), the highlighted cells should be deleted or troubleshooted further. This will correct the sBRS calculation. Additionally, while the midpoint DBP of each bin is presented in this calculation, in reality, using the floor, midpoint, or ceiling of each bin for each X-value does not matter because it will not influence the slope or the correlation coefficient if they are all shifted equally. The second figure below shows what happens when the macro button is pressed and the bin number is selected (Bin #1 in this example). Easily find in Column H the erroneous blood pressure value and decide whether or not to delete this value.



SYMPATHETIC BAROREFLEX SENSITIVITY CALCULATIONS

DBP Floor	-6	DBP Bin #	DBP Bins	# Beats/Bin	# Bursts/Bin	Burst Incidence/Bin	sBRS (unweighted)
DBP Ceiling	72	#1	-6	1	1	100	-0.517058214
Average DBP	58.76146705	#2	-3	0	0		
		#3	0	0	0		
		#4	3	0	0		
		#5	6	0	0		
		#6	9	0	0		
		#7	12	0	0		
		#8	15	0	0		
		#9	18	0	0		
		#10	21	0	0		
		#11	24	0	0		
		#12	27	0	0		
		#13	30	0	0		
		#14	33	0	0		
		#15	36	0	0		
		#16	39	0	0		
		#17	42	0	0		
		#18	45	0	0		
		#19	48	2	2	100	
		#20	51	32	10	31.25	
		#21	54	104	46	44.23076923	
		#22	57	160	84	52.5	
		#23	60	125	71	56.8	
		#24	63	61	37	60.6557377	



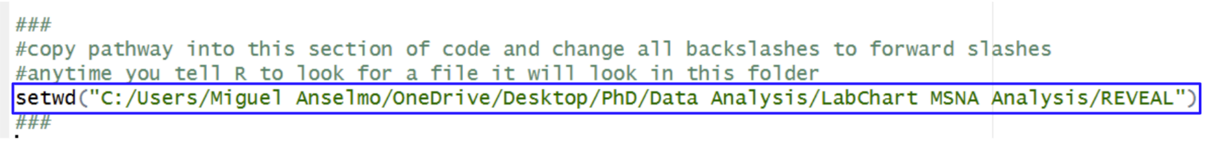
F	G	H	I
107.15	84.83	85.90	
106.14	84.84	85.90	
107.12	85.86	84.83	
107.12	85.86	84.84	
107.23	85.81	-6.00	
108.00	-6.00	85.87	
108.00	86.27	85.81	
108.22	86.90	86.27	
108.64	86.26	86.90	

6. When performing MSNA analysis for CPT or EPR (or other stressors), using the DATAPAD_BEAT macro may result in incorrect data exportation in individuals with very high heart rates. If erroneous BP data is noticed, then simply open the DATAPAD_BEAT macro, and shorten the RegionSecs time period from 0.25 to 0.15, or less, depending on the HR. This may require a bit of thinking and troubleshooting, and is very rare to encounter, but something to be aware of. This error may appear as repeated BP values across several cardiac cycles.

Compiling data using R studio script. Skip this part if choosing to not use the script to compile the data.

1. First, ensure that all Excel workbooks are in one folder under one file pathway.
2. Download R and R studio.
3. The only portions that must be changed are specified below.

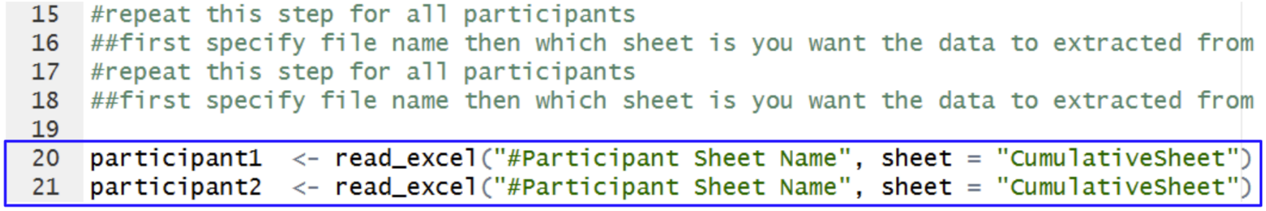
4.



```
###
#copy pathway into this section of code and change all backslashes to forward slashes
#anytime you tell R to look for a file it will look in this folder
setwd("C:/Users/Miguel Anselmo/OneDrive/Desktop/PhD/Data Analysis/LabChart MSNA Analysis/REVEAL")
###
```

Copy and paste the folder file pathway where the data exists. Next, **change all backslashes to forward slashes**.

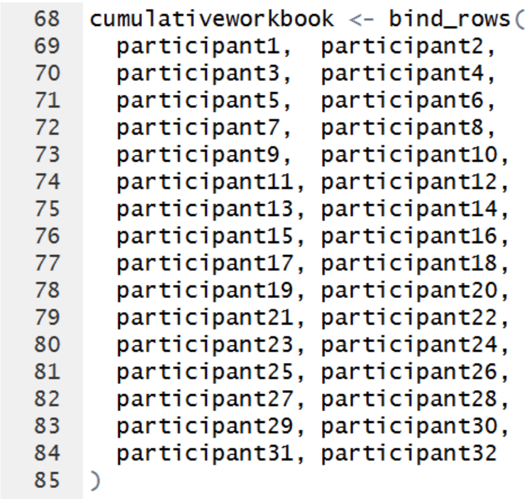
5.



```
#repeat this step for all participants
##first specify file name then which sheet is you want the data to extracted from
#repeat this step for all participants
##first specify file name then which sheet is you want the data to extracted from

participant1  <- read_excel("#Participant Sheet Name", sheet = "CumulativeSheet")
participant2  <- read_excel("#Participant Sheet Name", sheet = "CumulativeSheet")
```

Next, change the Participant Sheet Name section to each participant sheet that is available. Use the exact name that is specified in the Excel Folder.

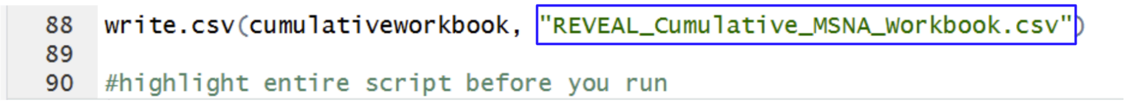
6.



```
cumulativeworkbook <- bind_rows(
  participant1,  participant2,
  participant3,  participant4,
  participant5,  participant6,
  participant7,  participant8,
  participant9,  participant10,
  participant11, participant12,
  participant13, participant14,
  participant15, participant16,
  participant17, participant18,
  participant19, participant20,
  participant21, participant22,
  participant23, participant24,
  participant25, participant26,
  participant27, participant28,
  participant29, participant30,
  participant31, participant32
)
```

Next, bind all participant files. This is an example of a dataset where there are 32 participant profiles. This and the previous step require some work, but there are methods of using AI sources to compile this code quickly.

7.



```
write.csv(cumulativeworkbook, "REVEAL_Cumulative_MSNA_Workbook.csv")

#highlight entire script before you run
```

Update the desired workbook name. Then, highlight the entire script before pressing “run.” The compiled workbook should now live in the same folder pathway where all other Excel files are housed.